

No. of Functional Features Included in the Solution

Date	3 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Dataset Loading	Loads applicant and credit datasets for analysis and model training.	Data Processing	Done	High
2	Data Preprocessing	Handles missing values, removes duplicates, and cleans the dataset.	Data Processing	Done	High
3	Feature Engineering	Encodes categorical variables and prepares model-ready features.	Data Processing	Done	High
4	Exploratory Data Analysis	Generates visualizations to understand applicant data.	EDA	Done	Medium
5	Machine Learning Model Training	Trains Logistic Regression, Decision Tree, Random Forest, and XGBoost models.	Machine Learning	Done	High
6	Model Evaluation	Compares models using Accuracy Score, Confusion Matrix, and Classification Report.	Machine Learning	Done	High
7	Flask Web Application	Provides a web interface for applicant data input and prediction.	Web Application	Done	High
8	IBM Cloud Deployment	Deploys the trained model for online real-time prediction.	Cloud Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Credit Card Prediction Form
2	Backend / Logic	4	Data Preprocessing, Feature Engineering, ML Prediction, Model Evaluation
3	Database / Storage	2	Applicant Dataset, Trained Model (.pkl)
4	API / Integration	2	Flask API, IBM Watson Machine Learning
5	Security / Authentication	—	